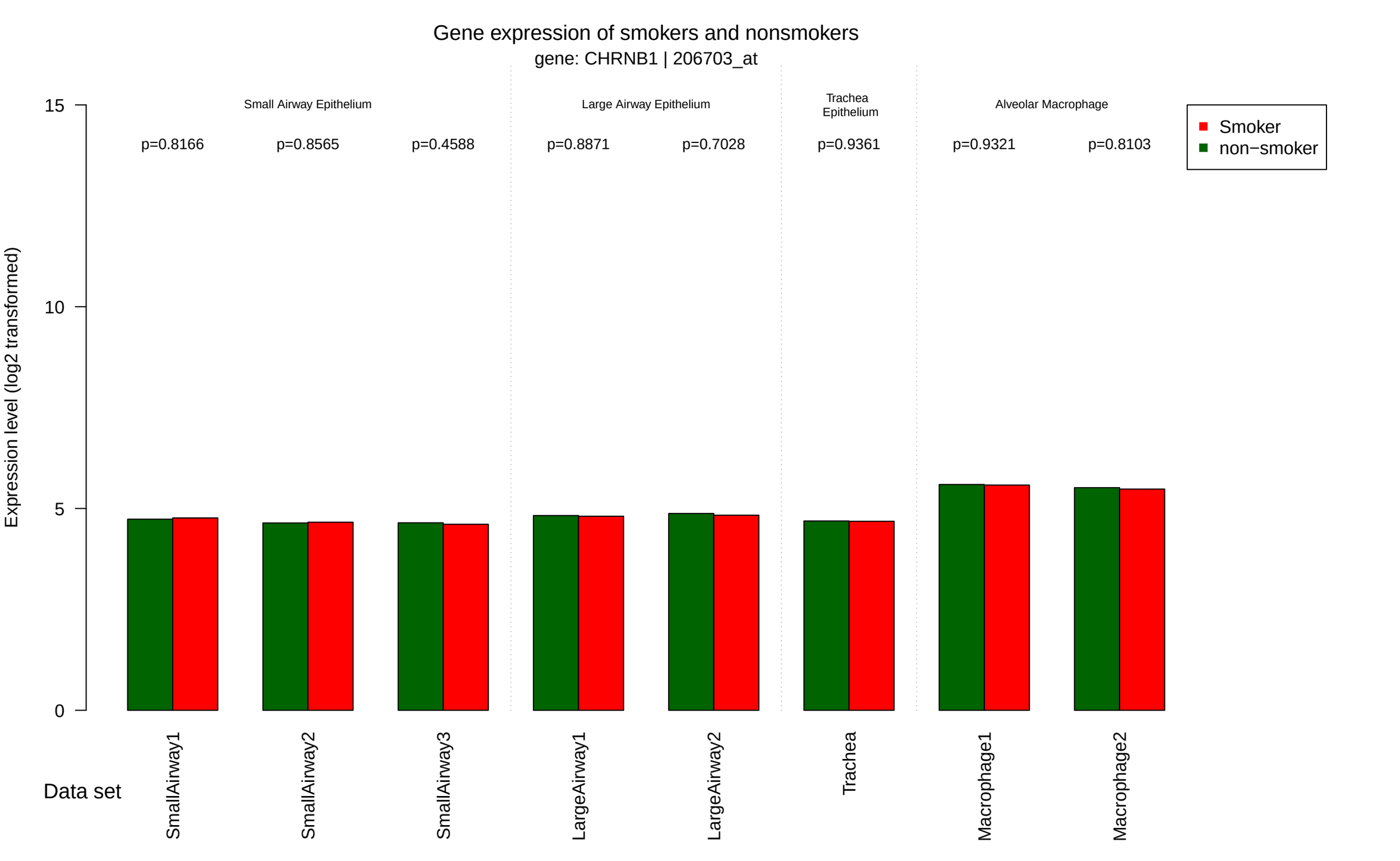
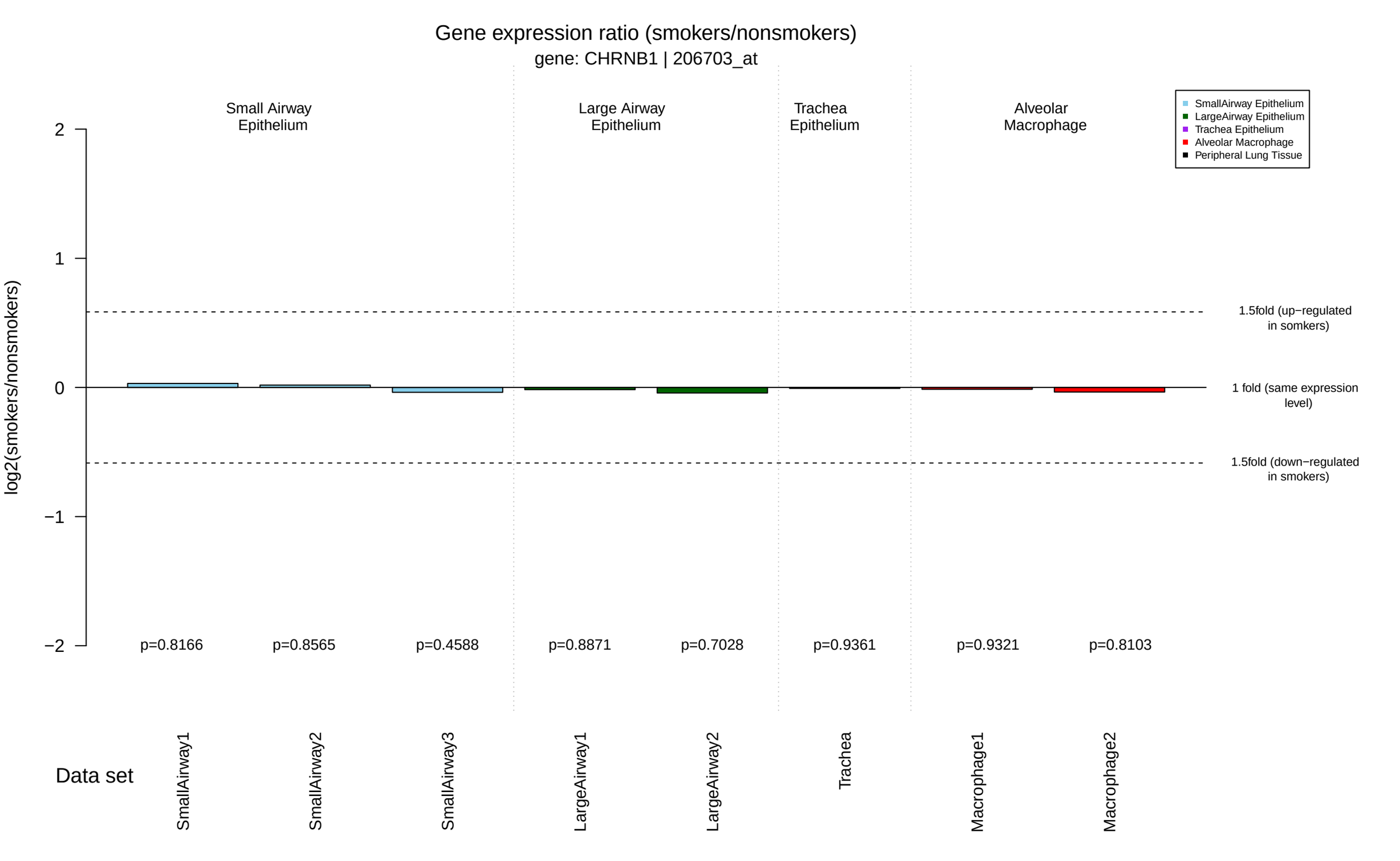
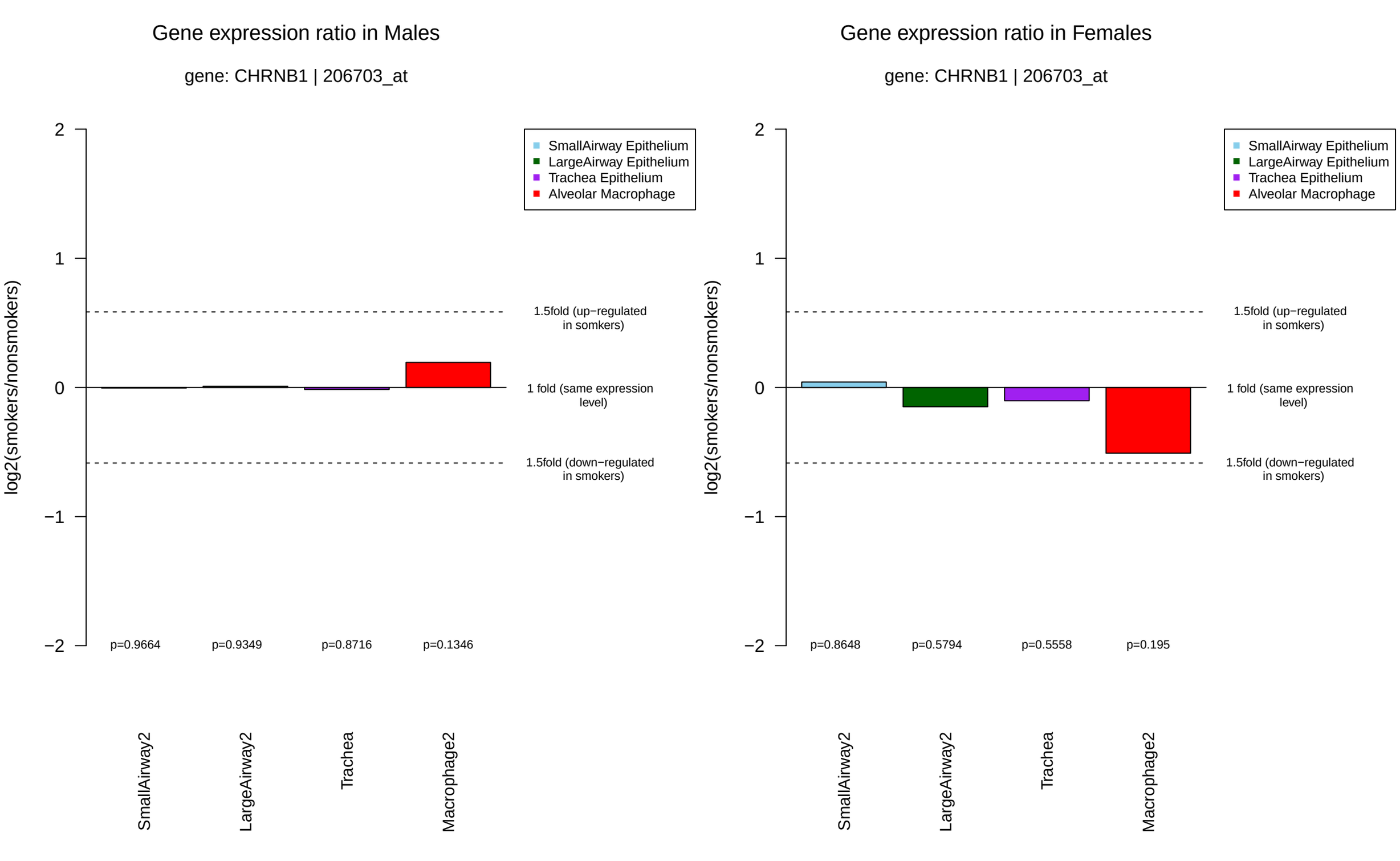
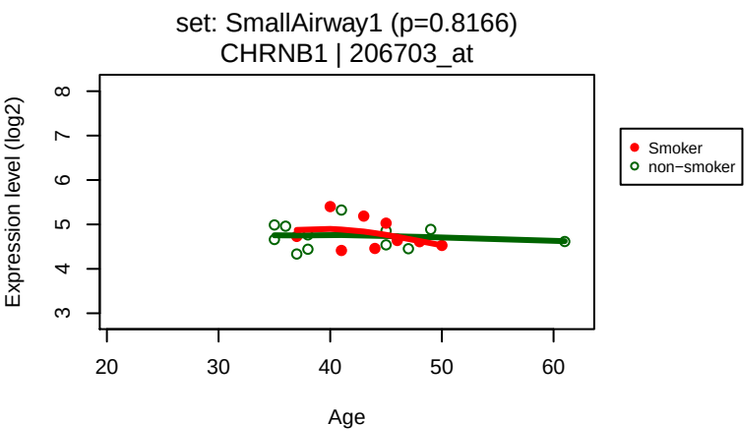
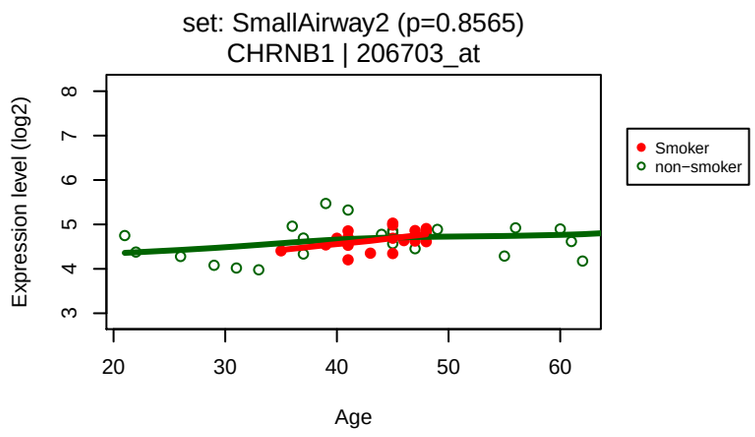
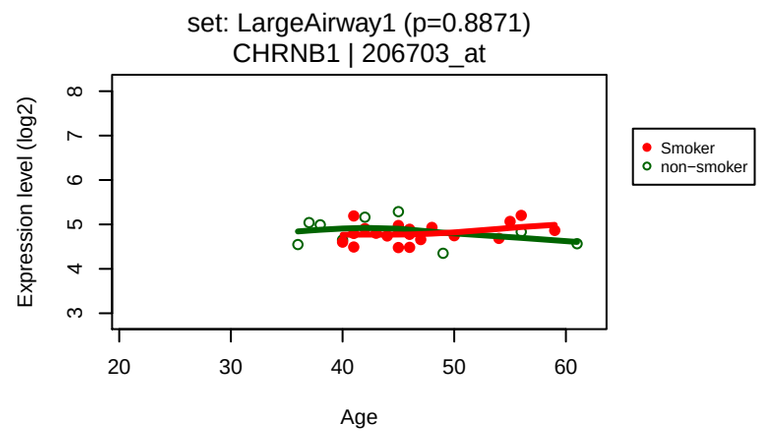

Gene expression of smokers and nonsmokers
gene: CHRNB1 | 206703_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8166
p=0.8565
p=0.4588
p=0.8871
p=0.7028
p=0.9361
p=0.9321
p=0.8103
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

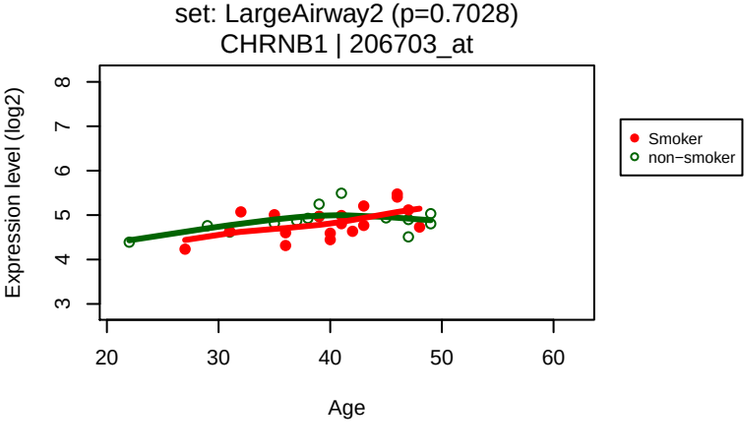

Gene expression ratio (smokers/nonsmokers)
gene: CHRNB1 | 206703_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8166
p=0.8565
p=0.4588
p=0.8871
p=0.7028
p=0.9361
p=0.9321
p=0.8103
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

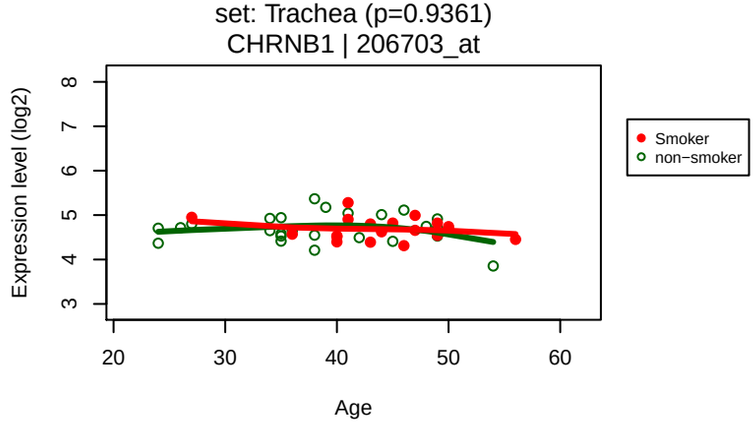

Gene expression ratio in Males
gene: CHRNB1 | 206703_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9664
p=0.9349
p=0.8716
p=0.1346
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHRNB1 | 206703_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8648
p=0.5794
p=0.5558
p=0.195
SmallAirway2
LargeAirway2
Trachea
Macrophage2

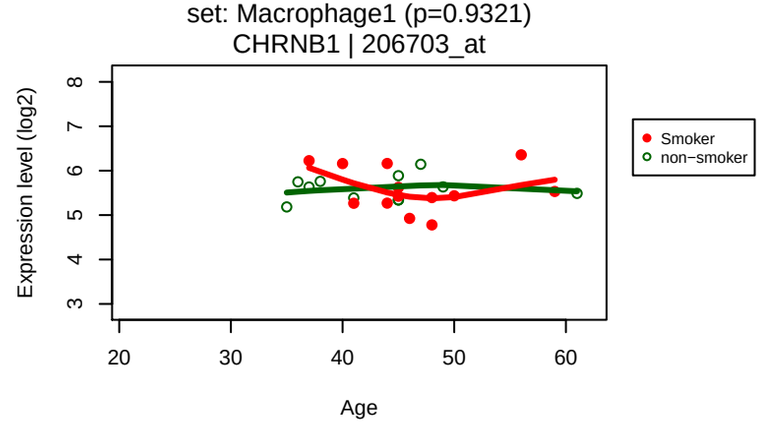

set: SmallAirway1 (p=0.8166)
CHRNB1 | 206703_at
Expression level (log2)
Age
Smoker
non−smoker

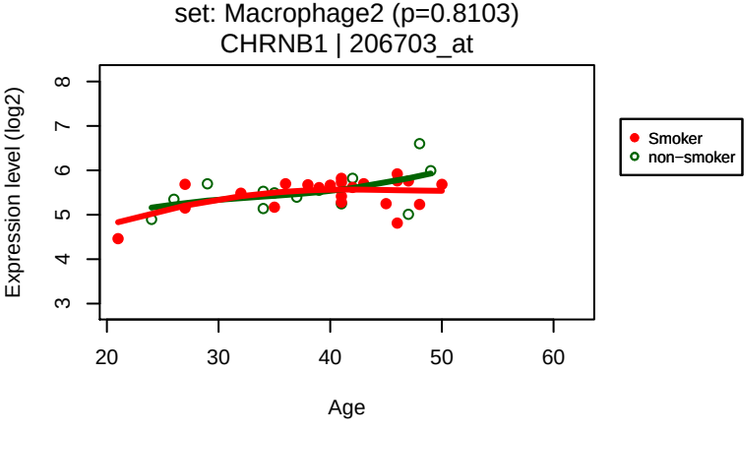

set: SmallAirway2 (p=0.8565)
CHRNB1 | 206703_at
Expression level (log2)
Age
Smoker
non−smoker

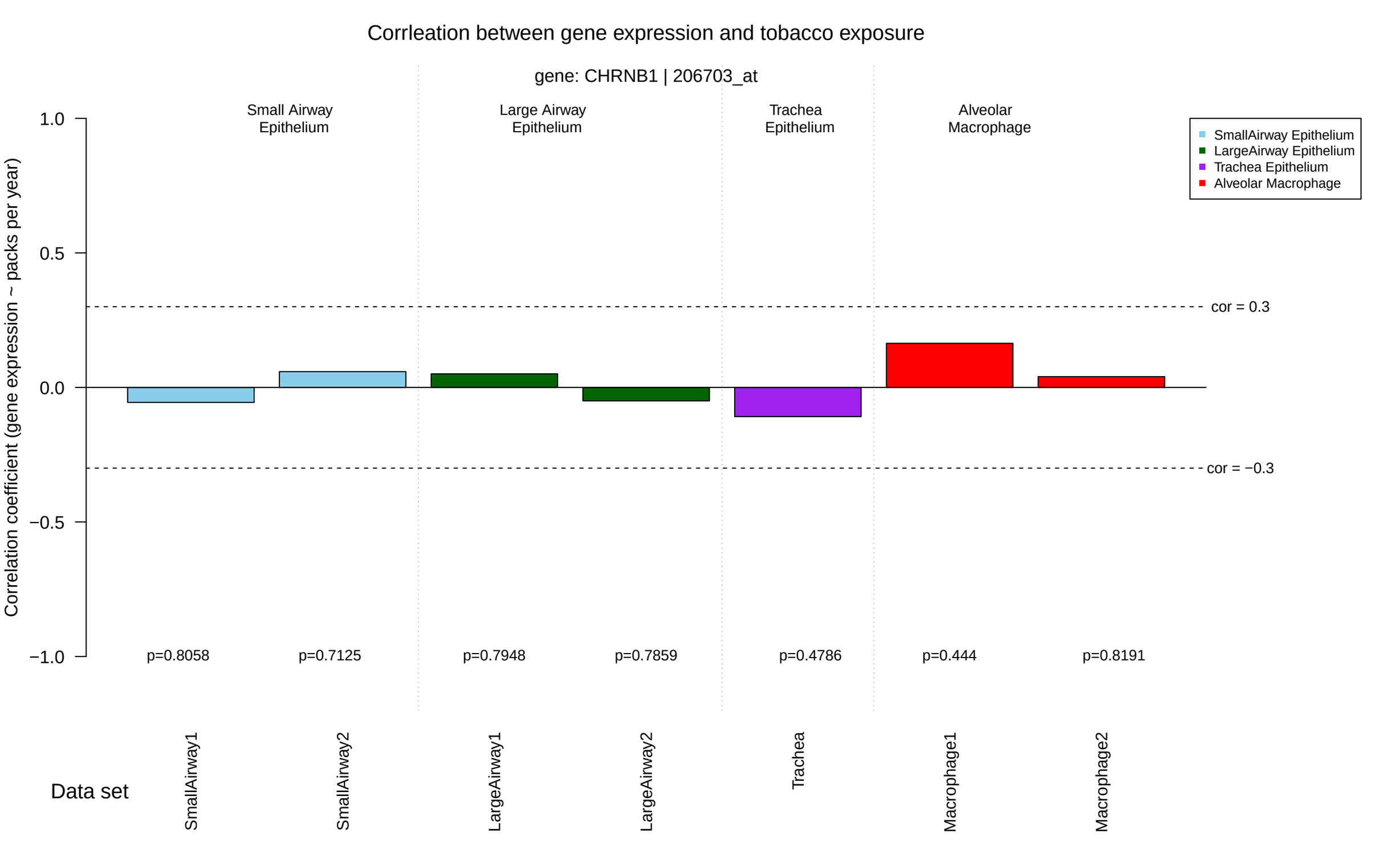

set: LargeAirway1 (p=0.8871)
CHRNB1 | 206703_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7028)
CHRNB1 | 206703_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9361)
CHRNB1 | 206703_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9321)
CHRNB1 | 206703_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8103)
CHRNB1 | 206703_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHRNB1 | 206703_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8058
p=0.7125
p=0.7948
p=0.7859
p=0.4786
p=0.444
p=0.8191
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2